

Conversion Between Base X and Base Y

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Conversion between a known base
and an unknown base.

$$34_7 = x_3$$

Convert the known base into decimal

$$34_7 = 3 \times 7^1 + 3 \times 7^0 = 25_{10}$$

Convert the decimal number into the unknown base

$$\frac{25}{3} = 8 \text{ (rem. 1)}, \frac{8}{3} = 2 \text{ (rem. 2)}, \frac{2}{3} = 0 \text{ (rem. 2)}$$

$$34_7 = a_2 a_1 a_0 = 221_3$$